



Learning Temporal, Relational, and Global Patterns in Multivariate Time Series for Accurate Anomaly Detection

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MINDS Lab

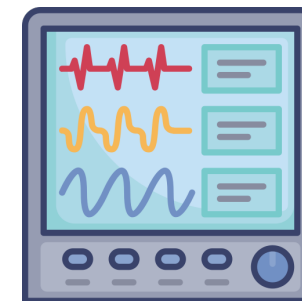
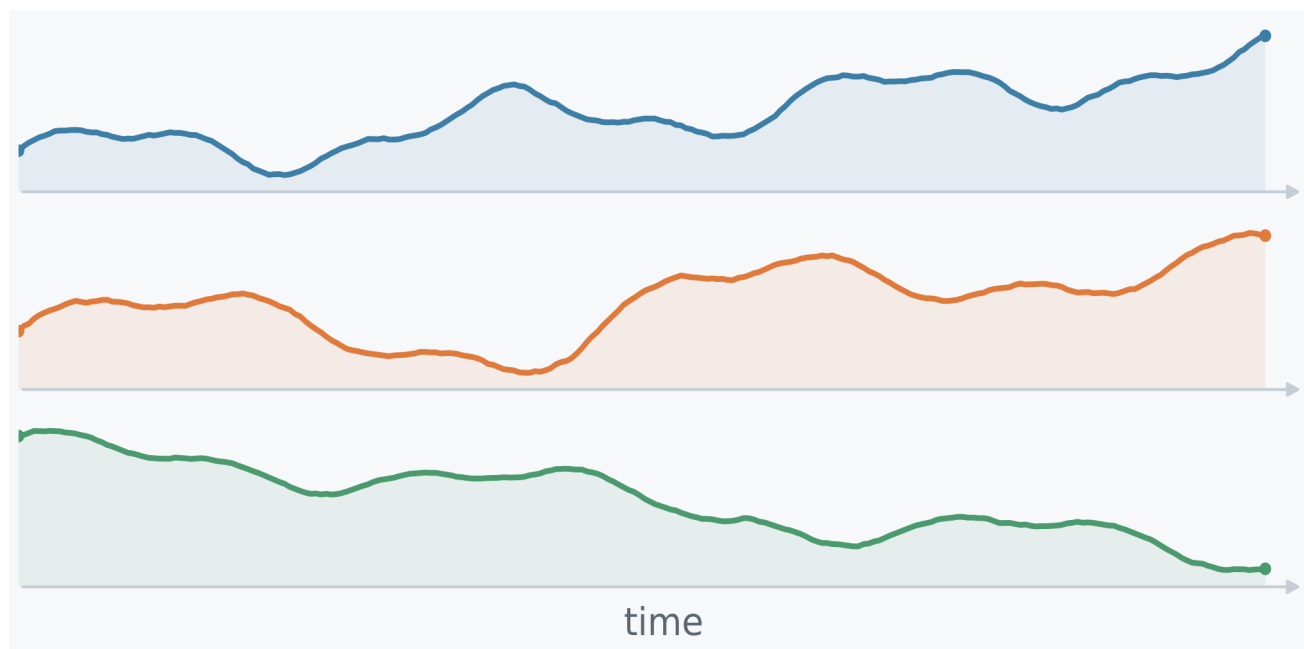
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- Time Series Anomaly Detection
- Motivation
- TIGER Framework
- Experimental Evaluation
- Conclusion

Time Series Anomaly Detection

■ What Is a Time Series?

- Values observed in temporal order



Healthcare

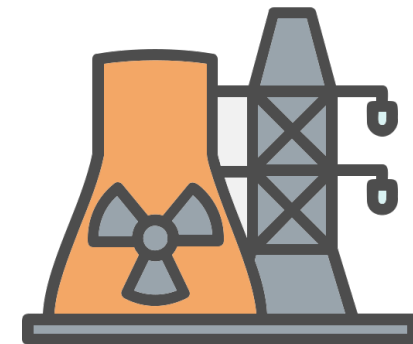
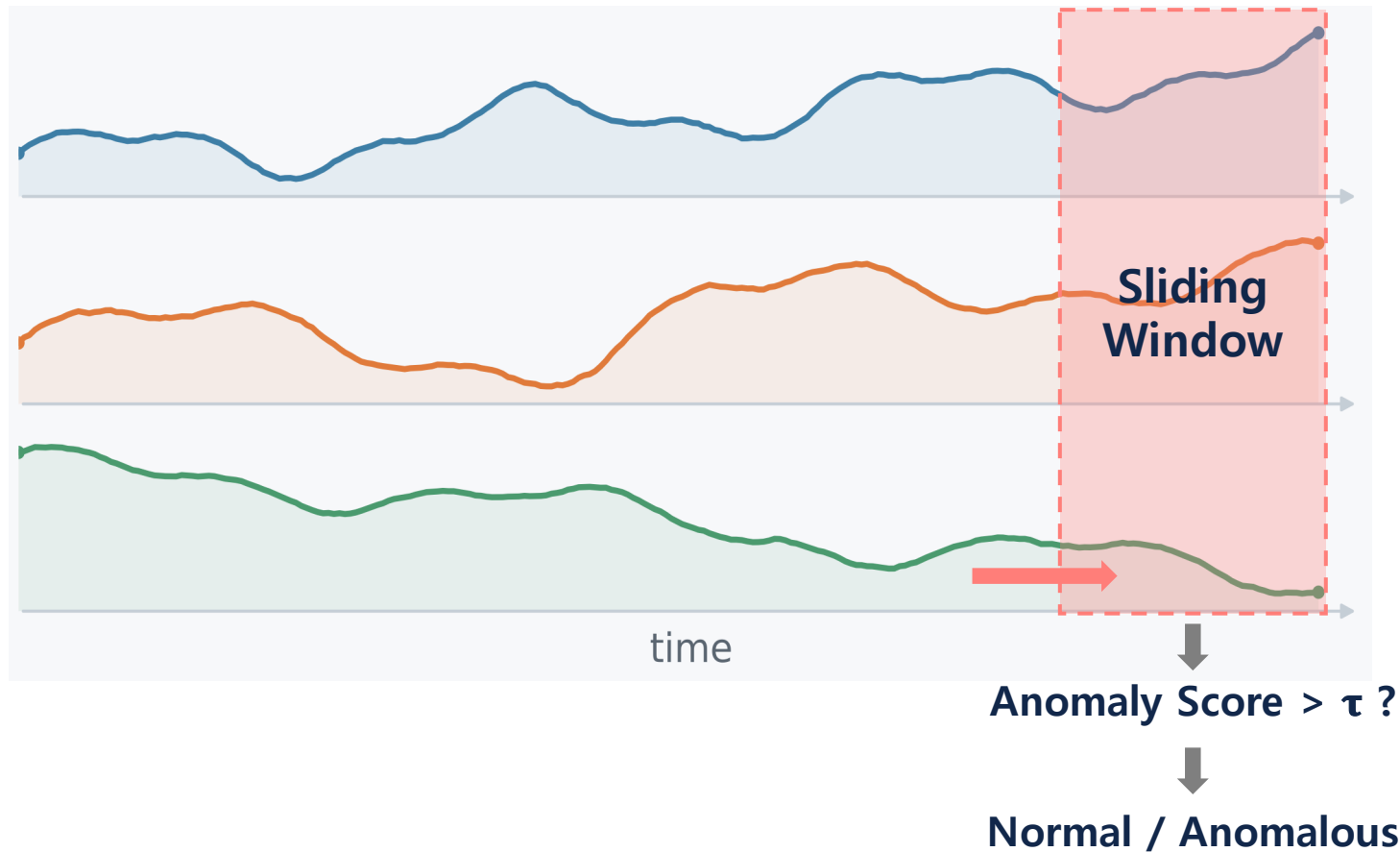


Industrial

Time Series Anomaly Detection

■ Task Overview

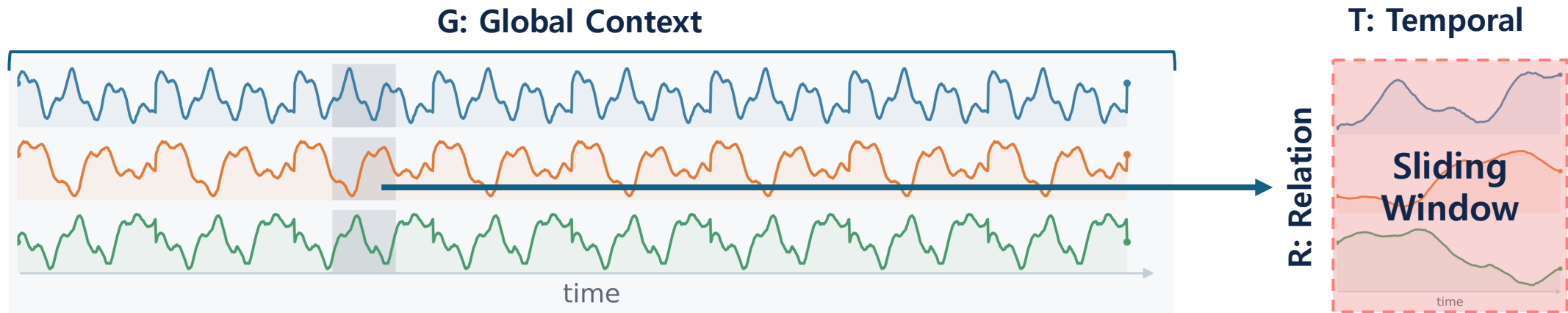
- Detect deviations from normal time-series behavior



Motivation

■ Three Complementary Information Axes

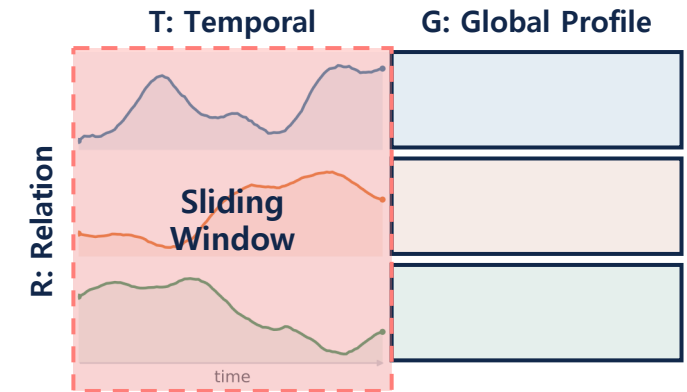
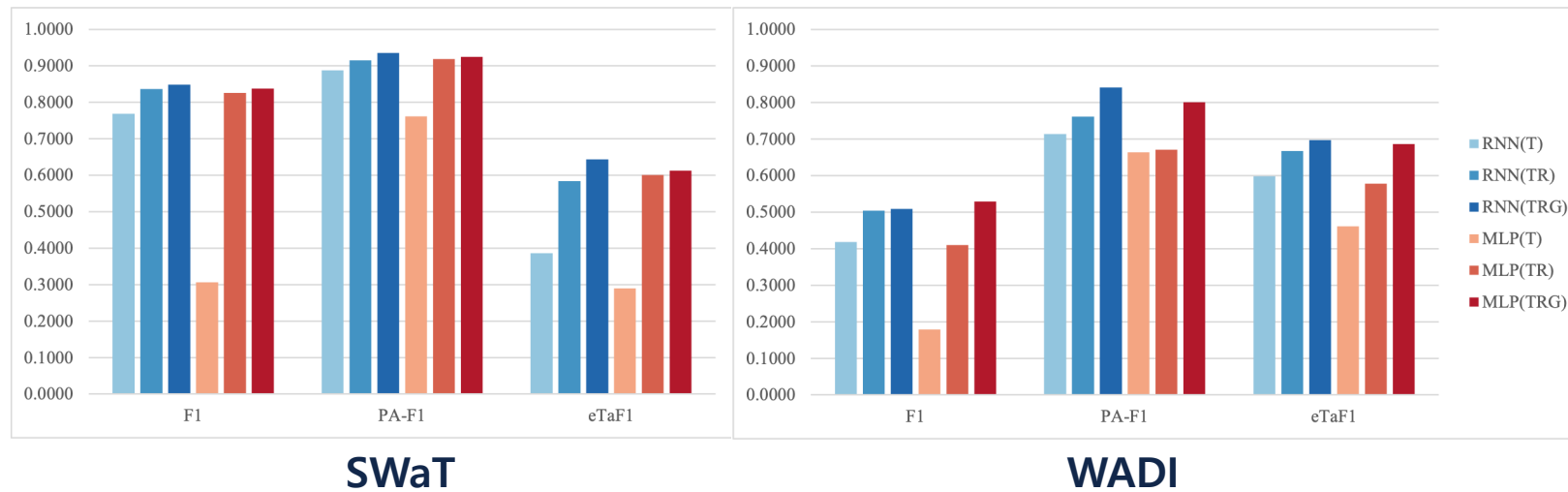
- ☐ (T) Intra-series Learning
- ☐ (R) Inter-series Learning
- ☐ (G) Global-context Learning



Motivation

■ Do These Perspectives Actually Matter?

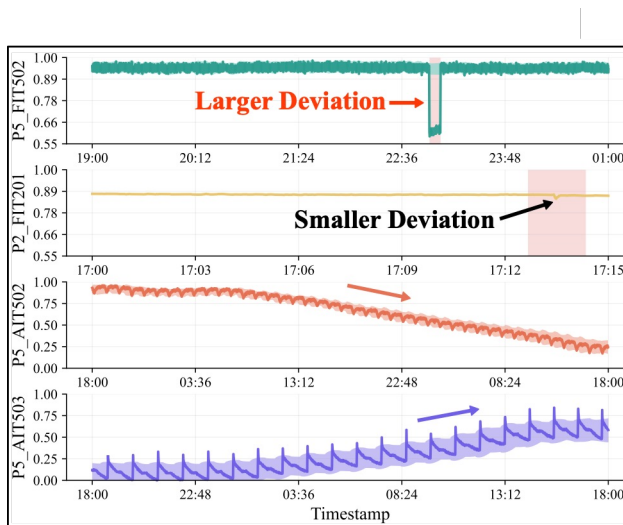
□ Simple Backbones MLP and RNN



→ Understanding normal behavior requires all three perspectives

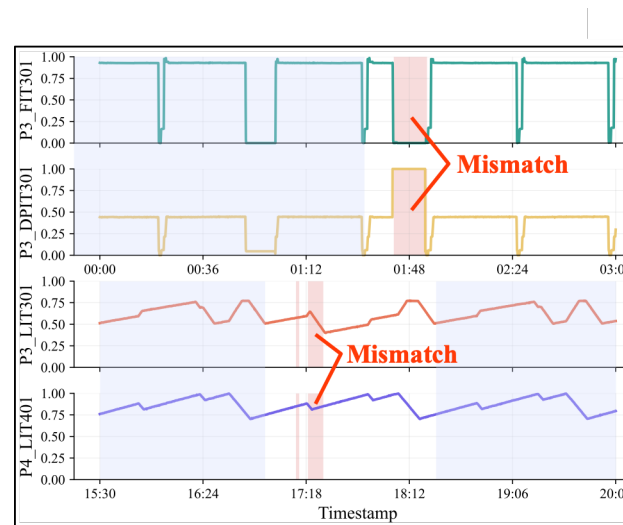
Motivation

■ What Are the Key Challenges in MTSAD?



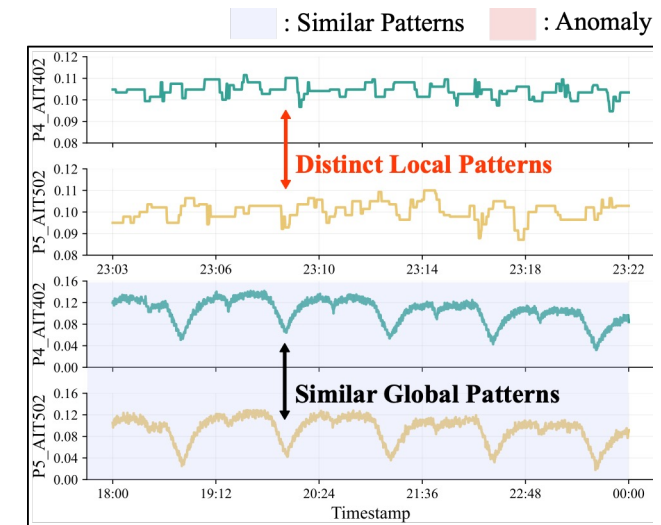
C1. Intra-series Learning

Normal patterns vary across series and temporal scales



C2. Inter-series Learning

Anomalies can break normal relationships



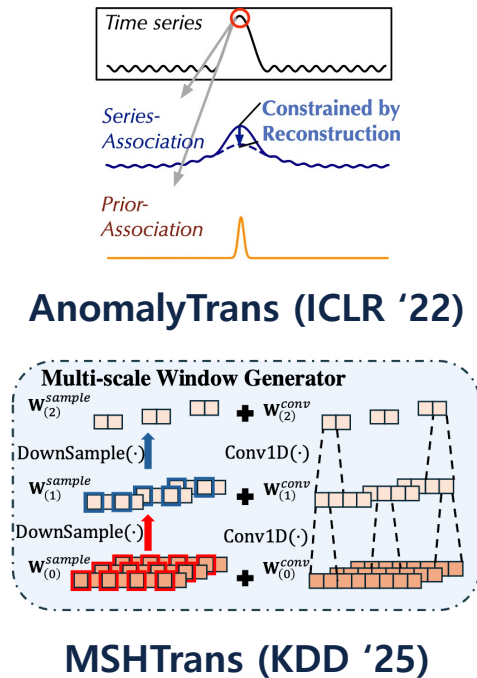
C3. Global-context Learning

Local patterns need long-term context

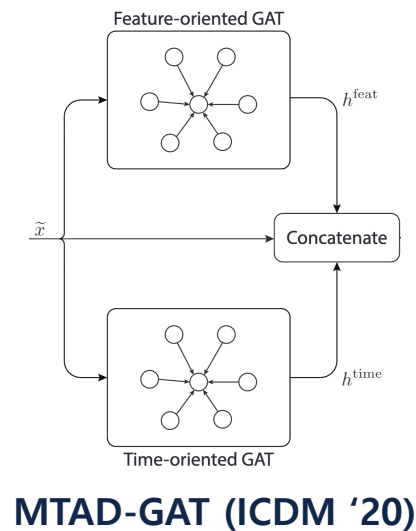
Motivation

■ Previous Work

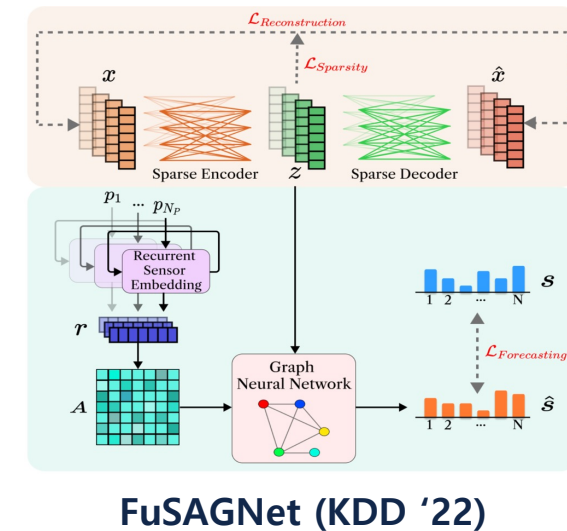
(T) Intra-series Learning



(R) Inter-series Learning



(G) Global-context Learning



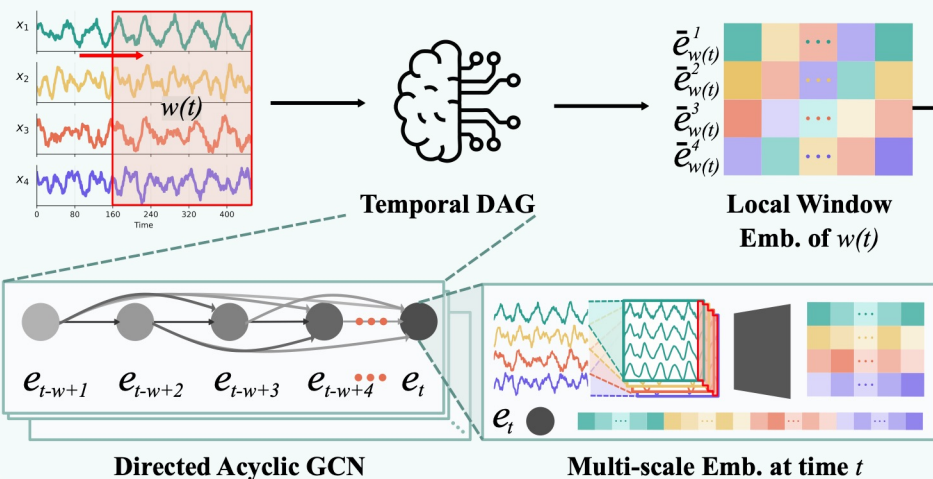
Goal: Unified learning of temporal, relational, and global representations

TIGER Framework

Integrating Temporal, Global, and Relational Information

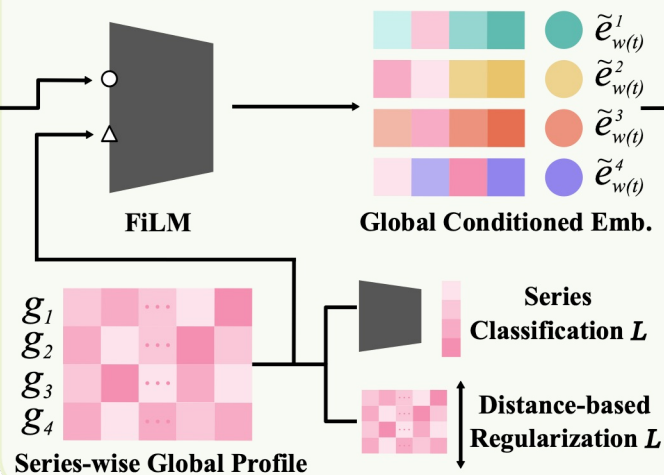
① (T) Intra-series Learning

Multi-scale DAG Encoder



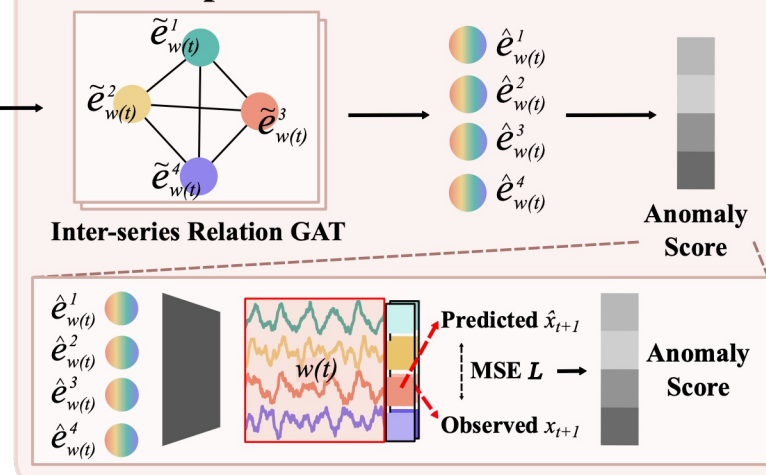
② (G) Global-context Learning

Series-wise Global Encoder

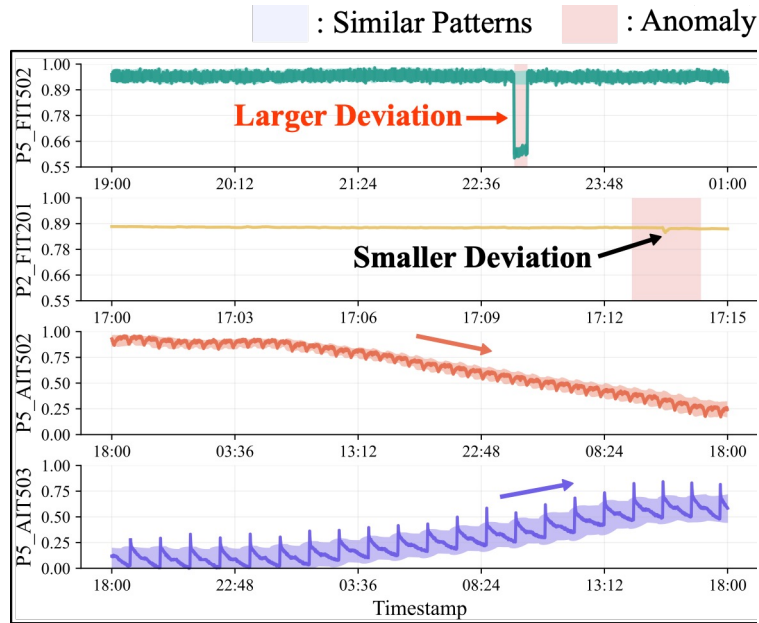


③ (R) Inter-series Learning

Graph-based Relation Encoder

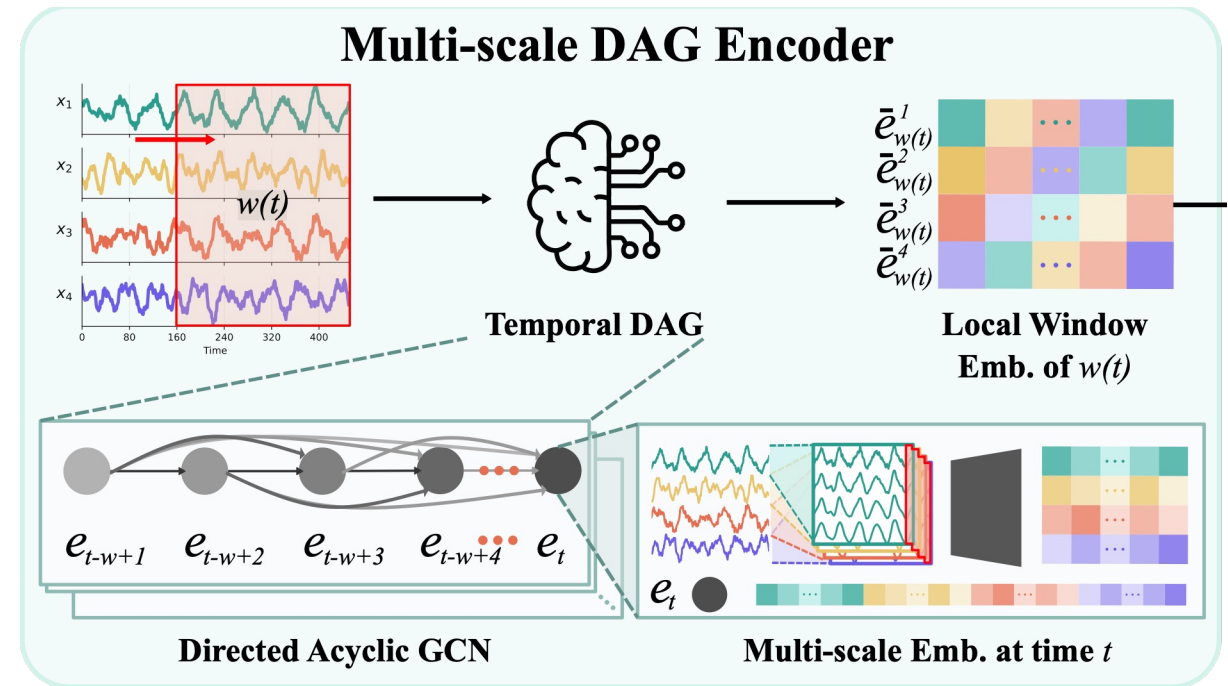


Multi-scale DAG Encoder



C1. Intra-series Learning

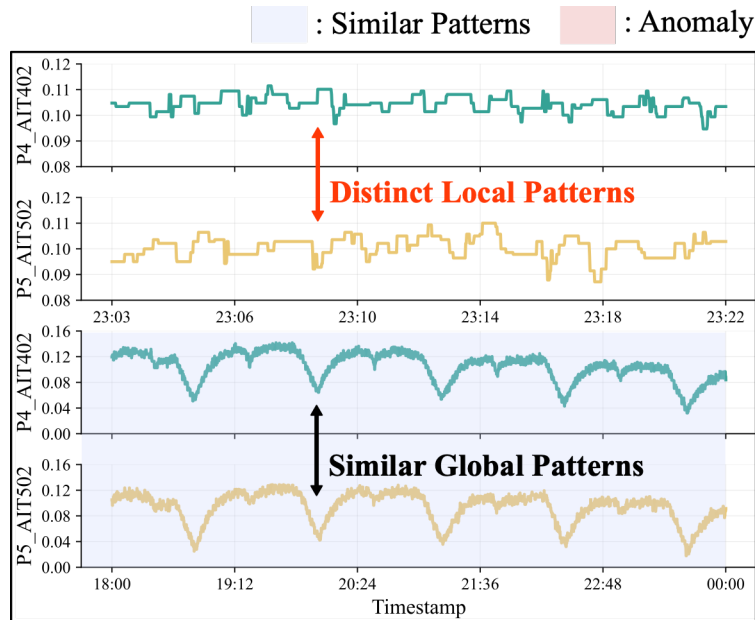
Normal patterns vary across series and temporal scales



Propagates information from past to present

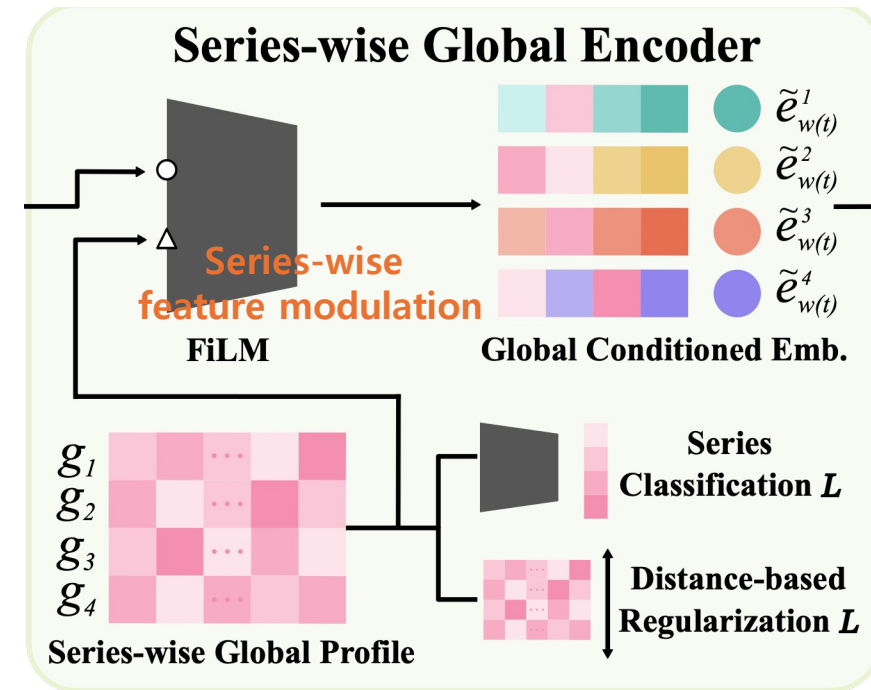
Captures patterns at multiple temporal scales

■ Series-wise Global Encoder



C3. Global-context Learning

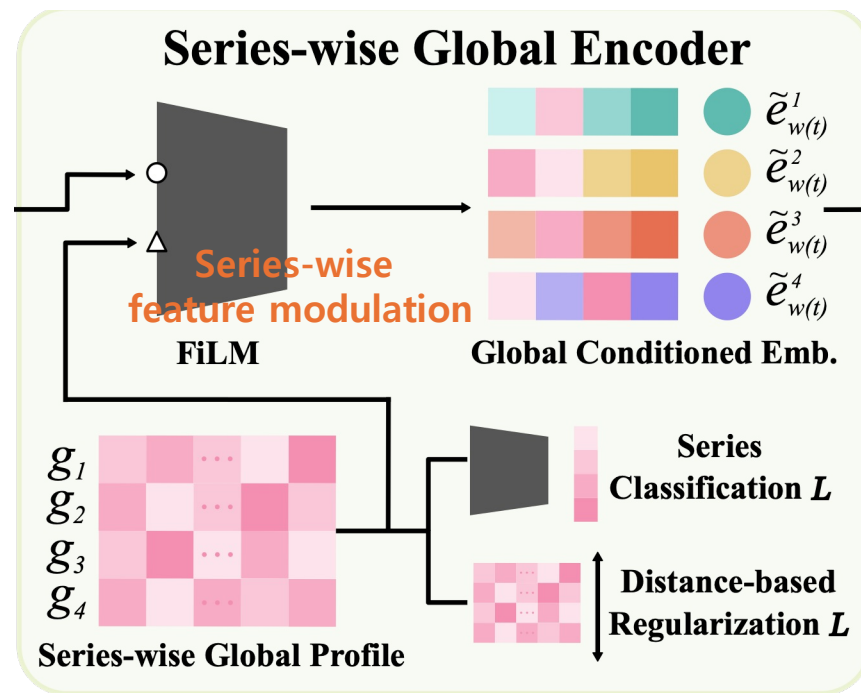
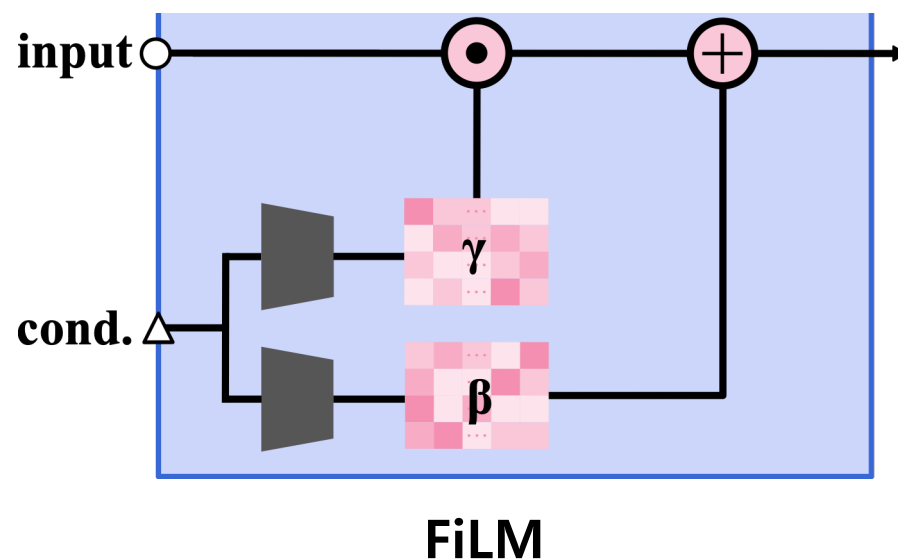
Local patterns need long-term context



Encourages distinct global profiles
for series-specific interpretation

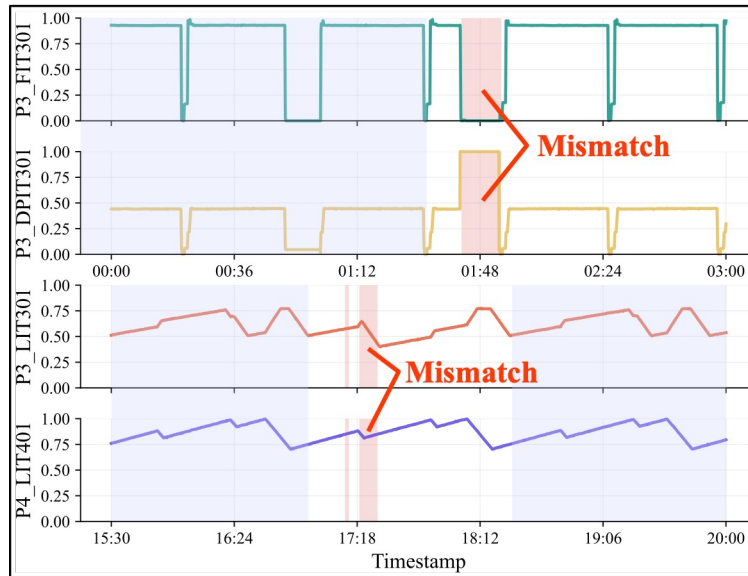
TIGER Framework

■ Series-wise Global Encoder



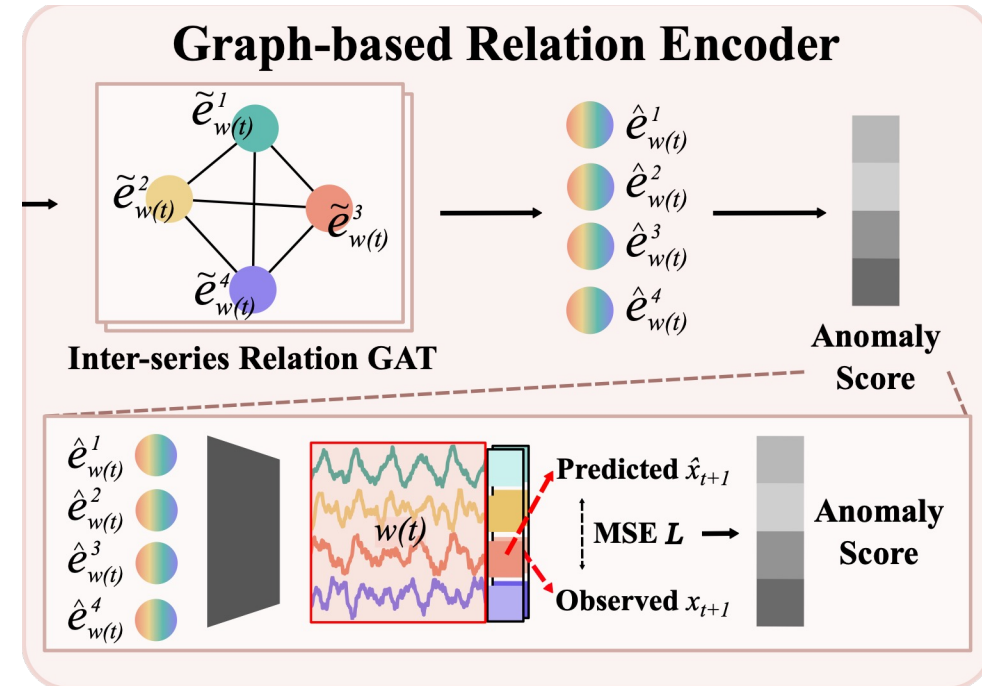
Encourages distinct global profiles
for series-specific interpretation

■ Graph-based Relation Encoder



C2. Inter-series Learning

Anomalies can break normal relationships



Refines each series representation
through attention-based message passing

■ Evaluation Setup

Evaluation Item	Setting
Datasets	SWaT, WADI
Thresholding	Fixed FPR = 0.002
Metrics	F1, PA-F1, eTaF1
Repeated Runs	5 random seeds

Dataset	#Process	#Series	#Train	#Test	%Anomaly
SWaT(2015)	6	51	47,340	44,993	12.14
WADI(2019)	4	127	76,299	17,281	5.77

Dataset

Information Axis	KNN	OCSVM	DeepSVDD	USAD	TSMixer	AnomalyT	FuSAGNet	MtsCID	MSHTrans	TIGER
Intra-series Learning	×	×	△	△	△	△	△	○	○	○
Inter-series Learning	×	×	△	△	○	△	○	○	△	○
Global-context Learning	×	×	×	×	×	×	△	×	×	○

Baseline

Experimental Evaluation

■ RQ1. Detection Performance

- ☐ Does TIGER works?

■ RQ2. Structural Design

- ☐ Which temporal and relational designs matter?

■ RQ3. Global Context

- ☐ Does it learn series-specific roles?

■ RQ4. Efficiency

- ☐ IS TIGER practical?

RQ1. Detection Performance

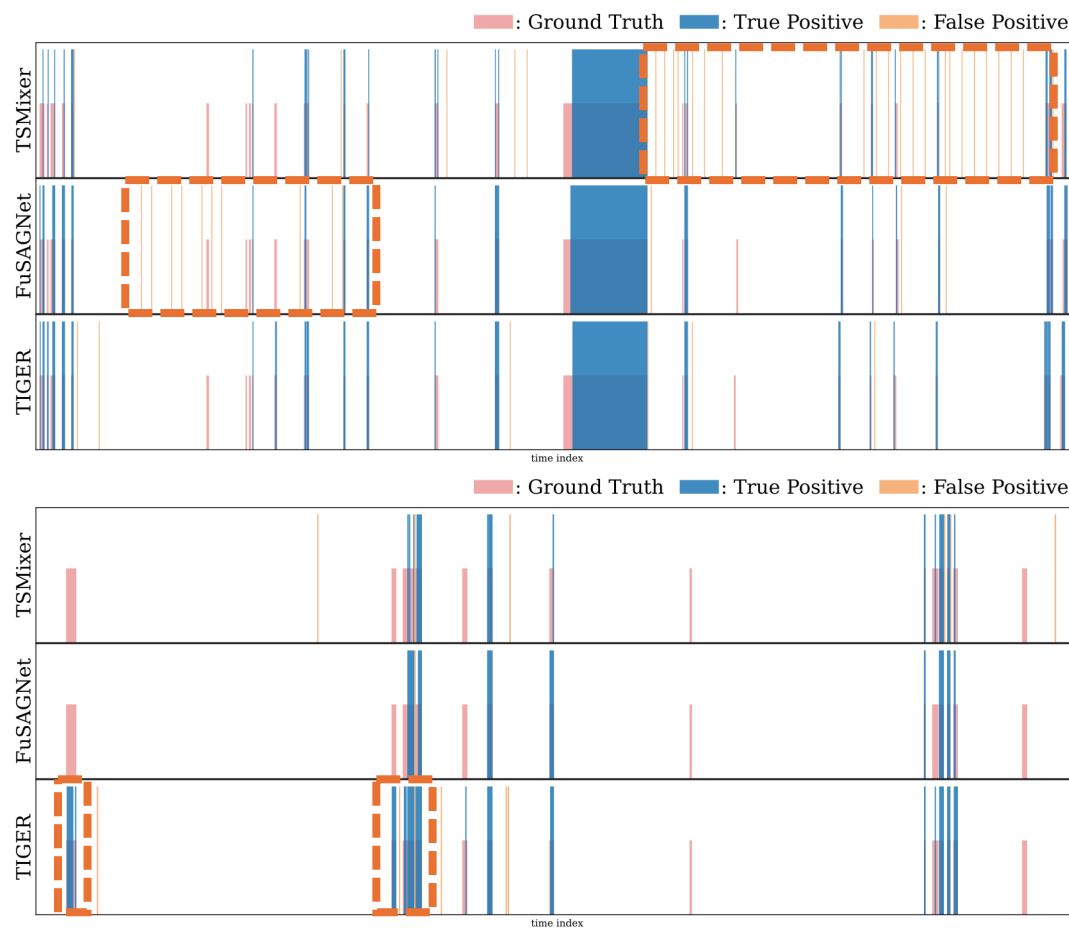
■ Does TIGER Works?

Dataset	Metric	KNN	OCSVM	DeepSVDD	USAD	TSMixer	AnomalyT	FuSAGNet	MtsCID	MSHTrans	TIGER
SWaT	Precision	0.9789	0.9788	0.9764	0.9810	0.9789	0.9789	<u>0.9814</u>	0.3436	0.9809	0.9824
	Recall	0.6526	0.6496	0.5829	0.7119	0.6534	0.6551	<u>0.7449</u>	0.0073	0.7154	0.7855
	F1	0.7831	0.7809	0.7300	0.8250	0.7836	0.7847	<u>0.8469</u>	0.0143	0.8273	0.8729
	PA-Precision	0.9814	0.9814	0.9795	0.9833	0.9847	0.9822	<u>0.9843</u>	0.9814	0.9838	0.9847
	PA-Recall	0.7435	0.7435	0.6706	0.8106	0.9085	0.7812	<u>0.8819</u>	0.7375	0.8458	<u>0.9050</u>
	PA-F1	0.8461	0.8461	0.7961	0.8884	0.9451	0.8699	0.9302	0.8421	0.9095	<u>0.9431</u>
	eTaP	0.9099	0.9346	<u>0.9228</u>	0.9162	0.8086	0.8339	0.8872	0.3423	0.8744	0.9126
	eTaR	0.2411	0.2400	<u>0.0483</u>	0.3291	0.4625	0.2778	<u>0.4915</u>	0.1077	0.4027	0.5881
	eTaF1	0.3812	0.3819	0.0917	0.4831	0.5878	0.4146	<u>0.6319</u>	0.1635	0.5500	0.7145
	Precision	0.8137	0.8137	0.1629	0.8998	0.8396	0.7958	<u>0.9289</u>	0.1053	0.8481	0.9520
WADI	Recall	0.1314	0.1314	0.0086	0.2658	0.1727	0.1338	<u>0.3823</u>	0.0036	<u>0.3994</u>	0.5805
	F1	0.2263	0.2263	0.0164	0.4101	0.2829	0.2267	<u>0.5414</u>	0.0070	<u>0.5406</u>	0.7202
	PA-Precision	0.8429	0.8429	0.2735	<u>0.9616</u>	0.9495	0.9298	0.9574	0.8631	0.9093	0.9684
	PA-Recall	0.1615	0.1615	0.0269	<u>0.7476</u>	0.5665	0.4090	0.6802	0.1968	0.7184	0.8985
	PA-F1	0.2710	0.2710	0.0489	<u>0.8386</u>	0.7058	0.5655	0.7891	0.3196	0.7978	0.9307
	eTaP	0.4650	0.4705	0.1061	0.8400	0.7341	0.7142	<u>0.8707</u>	0.1006	0.7365	0.8934
	eTaR	0.1961	0.1961	0.0189	0.4666	0.3902	0.2750	<u>0.5534</u>	0.1092	<u>0.5582</u>	0.7161
	eTaF1	0.2759	0.2768	0.0319	0.5990	0.5063	0.3969	<u>0.6733</u>	0.1045	<u>0.6311</u>	0.7944
Avg. rank ↓		7.50	7.33	9.67	4.00	4.17	5.83	<u>2.50</u>	9.00	3.33	1.17

RQ1. Detection Performance

■ Does TIGER Works?

Dataset	Metric	TSMixer	FuSAGNet	TIGER
SWaT	Precision	0.9789	<u>0.9814</u>	0.9824
	Recall	0.6534	<u>0.7449</u>	0.7855
	F1	0.7836	<u>0.8469</u>	0.8729
	PA-Precision	0.9847	<u>0.9843</u>	0.9847
	PA-Recall	0.9085	0.8819	<u>0.9050</u>
	PA-F1	0.9451	0.9302	<u>0.9431</u>
	eTaP	0.8086	0.8872	0.9126
	eTaR	0.4625	<u>0.4915</u>	0.5881
	eTaF1	0.5878	<u>0.6319</u>	0.7145
WADI	Precision	0.8396	<u>0.9289</u>	0.9520
	Recall	0.1727	0.3823	0.5805
	F1	0.2829	<u>0.5414</u>	0.7202
	PA-Precision	0.9495	0.9574	0.9684
	PA-Recall	0.5665	0.6802	0.8985
	PA-F1	0.7058	0.7891	0.9307
	eTaP	0.7341	<u>0.8707</u>	0.8934
	eTaR	0.3902	<u>0.5534</u>	0.7161
	eTaF1	0.5063	<u>0.6733</u>	0.7944
Avg. rank ↓		4.17	<u>2.50</u>	1.17



RQ2. Structural Design

■ Which Temporal and Relational Designs Matter?

Model	Ablation		SWaT			WADI		
	Multi-Scale	DAG	F1	PA-F1	eTaF1	F1	PA-F1	eTaF1
TIGER	O	O	0.8729	0.9431	0.7145	0.7202	0.9307	0.7944
TIGER_SS	X	O	<u>0.8649</u> (−0.92%)	<u>0.9365</u> (−0.70%)	<u>0.6864</u> (−3.93%)	<u>0.5968</u> (−17.13%)	<u>0.9144</u> (−1.75%)	<u>0.7529</u> (− 5.22%)
TIGER_FC	O	X	0.8626(−1.18%)	0.9311(−1.27%)	0.6488(−9.20%)	0.5687(−21.04%)	0.8401(−9.73%)	0.7094(−10.70%)

Model	Ablation		SWaT			WADI		
	Inter-Series		F1	PA-F1	eTaF1	F1	PA-F1	eTaF1
TIGER	GAT		0.8729	0.9431	0.7145	0.7202	0.9307	0.7944
TIGER	GCN		0.8449(−3.21%)	0.9337(−1.00%)	0.6337(−11.31%)	0.5487(−23.81%)	0.8487(−8.81%)	0.7103(−10.58%)
TIGER	MLP		<u>0.8563</u> (−1.90%)	<u>0.9428</u> (−0.03%)	<u>0.6762</u> (− 5.36%)	<u>0.6710</u> (− 6.83%)	<u>0.8870</u> (−4.70%)	<u>0.7423</u> (− 6.56%)

RQ3. Global Context

■ Does It Learn Series-specific Roles?

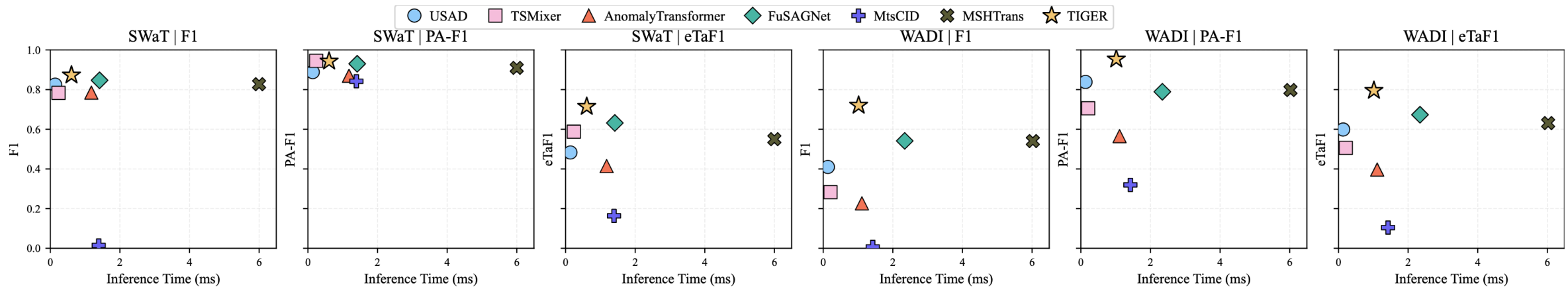
Model	Ablation		SWaT			WADI		
	Method	Granularity	F1	PA-F1	eTaF1	F1	PA-F1	eTaF1
TIGER	X	X	0.8153(−6.60%)	0.8850(−6.16%)	0.5510(−22.88%)	0.3523(−51.08%)	0.4263(−54.20%)	0.4574(−42.42%)
TIGER	Concat	Single	0.8519(−2.41%)	0.9252(−1.90%)	0.6469(− 9.46%)	<u>0.6372</u> (−11.52%)	<u>0.9280</u> (− 0.29%)	<u>0.7590</u> (− 4.46%)
TIGER	Concat	Process-wise	0.8493(−2.70%)	0.9278(−1.62%)	0.6398(−10.45%)	<u>0.6156</u> (−14.52%)	<u>0.8855</u> (− 4.86%)	<u>0.7488</u> (− 5.74%)
TIGER	Concat	Series-wise	0.8612(−1.34%)	0.9285(−1.55%)	0.6644(− 7.01%)	0.6016(−16.47%)	0.9077(− 2.47%)	0.7385(− 7.04%)
TIGER	FiLM	Single	<u>0.8694</u> (−0.40%)	0.9367(−0.68%)	<u>0.6654</u> (− 6.87%)	0.5871(−18.48%)	0.8856(− 4.85%)	0.7447(− 6.26%)
TIGER	FiLM	Process-wise	0.8651(−0.89%)	0.9437 (+0.06%)	0.6619(− 7.36%)	0.6287(−12.70%)	0.8962(− 3.71%)	0.7427(− 6.51%)
TIGER	FiLM	Series-wise	0.8729	<u>0.9431</u>	0.7145	0.7202	0.9307	0.7944

Model	Ablation		SWaT			WADI		
	Euc.	Cls.	F1	PA-F1	eTaF1	F1	PA-F1	eTaF1
TIGER	O	O	0.8729	0.9431	0.7145	0.7202	0.9307	0.7944
TIGER w/o Euc.	X	O	0.8584(−1.66%)	0.9286(−1.54%)	0.6194(−13.31%)	0.5852(−18.74%)	0.8926(−4.09%)	0.7235(−8.92%)
TIGER w/o Cls.	O	X	0.8629(−1.15%)	<u>0.9421</u> (−0.11%)	<u>0.6771</u> (− 5.23%)	<u>0.6451</u> (−10.43%)	0.8986(−3.45%)	<u>0.7606</u> (−4.25%)
TIGER w/o Aux.	X	X	<u>0.8636</u> (−1.07%)	0.9316(−1.22%)	0.6560(− 8.19%)	0.5880(−18.36%)	<u>0.9299</u> (−0.09%)	0.7364(−7.30%)

RQ4. Efficiency

■ IS TIGER Practical?

- ☐ All models satisfy practical real-time requirements
- ☐ Favorable accuracy–latency trade-off with strong detection performance



Conclusion

■ Three Complementary Perspectives

- Normal behavior requires temporal, relational, and global information

■ TIGER Framework

- Integrates multi-scale temporal patterns, series-wise global context, and inter-series relationships

■ Effective and Practical

- Consistent detection improvements with a favorable accuracy–latency trade-off

Q&A